

Assignment - 3 CSCV

Section - A

Q.1 IOT security refers to protecting Internet of Things devices and network from unauthorized access and attacks.

Two vulnerabilities :
• Weak / default password
• Lack of regular software updates.

Q.2 • Data interception (hackers can steal sensitive information)
• Man-in-the-Middle (MITM) attacks.

Q.3 Remote working security involves protecting data and systems when employees work outside the office.

Two threats :
• Phishing attacks
• Insecure networks

Q.4 Data disposal is the process of permanently deleting or destroying data.

Importance :
• Prevents data theft.
• Ensure privacy and compliance.

Q.5 Multi-Factor Authentication (MFA) requires multiple verification methods (e.g., password + OTP). It improves security by adding an extra layer beyond passwords.

Section - B

Q.1 IoT Devices Security Risks

Risks :

- Unauthorized access
- Data leak
- Botnet attacks

Attack Methods :

- Exploiting weak passwords
- Malware injection
- Network sniffing

Countermeasures :

- Use strong passwords
- Regular firmware updates
- Network segmentation
- Enable encryption

Q.2 Public Wi-fi & Remote Access

Risks :

- Data theft
- Session hijacking

Attack Methods :

- Man-in-the-Middle
- Rogue hotspots
- Packet sniffing

Best Practices :

- Use VPN
- Avoid accessing sensitive data
- Enable HTTPS
- Use MFA

Q.3

Secure Data Disposal

Risk of Improper Disposal :

- Identity theft
- Data breaches
- Legal issues

Techniques :

- Data wiping (software - based)
- Decryption + deletion
- Physical destruction (shredding drives)

Section - C

Q.1

Components :

- Device Security : Strong authentication, firmware updates
- Network Security : Firewalls, encryption of Wi-Fi.
- Data protection : Encryption and secure storage.

Threat Mitigation :

- Intrusion detection systems
- Disable unused features
- Regular monitoring

Privacy Protection :

- Limit data sharing
- Use trusted vendors
- Control permissions

Q.2

Risks :

- Phishing and malware
- Insecure devices
- Data leakage

Secure Communication Practices : • Use VPN and encrypted channels.
• Secure email systems
• Avoid public Wi-Fi.

Guidelines : • Use strong passwords
• Enable MFA
• Keep software updated
• Follow company policies.

Q.3 Secure Data Deletion & SOP

Technique : • Data wiping (multiple overwrite)
• Degaussing
• Physical destruction

Standard Operating Procedure (SOP):

1. Backup important data.
2. Use certified wiping tools.
3. Verify data removal
4. Destroy storage if needed
5. Maintain disposal records.